Information Theory Mn908 and MICAS

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Problem 1 ***The Binary Jammer Channel***

1. If the encoder and decoder do not know when the channel is blocked, the interference simply increases the channel crossover probability. If the channel is free, which occurs with probability q, a transmitted 0 is received as a 1 with probability ϵ. If the channel is blocked, which occurs with probability $1-q$, a transmitted 0 is received as a 1 with probability 1/2. The combined probability that a 0 is received as a 1 is therefore $q\epsilon + \frac{1-q}{2}$, and the channel capacity is
$$C = 1 - H_{\mathrm{b}}\left(q\epsilon + \frac{1-q}{2}\right). \tag{1}$$

2. In this case, a channel blockage is equivalent to an erasure event as it can be detected by the decoder and the encoder is not informed about it. The input of this channel is binary, $X \in \{0,1\}$, while the output is ternary, $Y \in \{0,1,\mathrm{E}\}$, for E denoting the blockage event. The channel transition law is
$$P_{Y|X}(y|x) = \begin{cases} q(1-\epsilon) & x = y \\ 1-q & y = \mathrm{E} \\ q\epsilon & \text{otherwise.} \end{cases}$$
To derive the channel capacity of this channel, let B be a binary random variable denoting the state of the channel, i.e., $B = 0$ corresponds to a free channel, and $B = 1$ corresponds to a blocked channel. Note that B is known to the receiver, but not to the transmitter. Thus,
$$\begin{aligned} C &= \max I(X;Y,B) && (2)\\ &= \max I(X;B) + I(X;Y|B) && (3)\\ &= \max I(X;Y|B) && (4)\\ &= \max\Big\{(1-q)I(X;Y|B=1) + qI(X;Y|B=0)\Big\} \\ &= \max qI(X;Y|B=0) && (5)\\ &= q(1-H_{\mathrm{b}}(\epsilon)) && (6) \end{aligned}$$
where (3) follows from the chain rule; (4) follows because the encoder does not know when the channel is blocked and hence X and B are independent of each other; equality in (5) holds because when the channel is blocked, then X and Y are independent; and (6) holds because the unblocked channel is a binary symmetric channel with crossover probability ϵ.

3. Let the message set be $\mathcal{W} = \{1,..,2^{nR}\}$ where $R = q(1-H(\epsilon))$. On average the channel will be blocked $(1-q)n$ times in n channel uses. As $n \to \infty$ the probability that the number of blocked channel uses is greater than $(1-q)n$ goes to zero. This follows from the law of large numbers. Therefore we will only get to use the "good" channel q proportion of the time.

   Let the encoder encode for the BSC with crossover probability ϵ. Then whenever the encoder observes a blocked channel it simply transmits a zero. Otherwise it transmits the next bit in

the encoding of w. The decoder can likewise observe the channel. If the channel is blocked then the decoder will ignore the bit it receives.

Thus effectively we only use the channel when the channel is not blocked. We know we can code arbitrarily close to capacity for large blocklengths. And we are coding for the BSC with capacity $1 - H(\epsilon)$. Thus the rate of this encoding scheme, $q(1 - H(\epsilon))$, is achievable.

Though it takes some effort to prove, one can show that the capacity remains the same even if the encoder knows the channel blockage locations beforehand. For most channels foreknowledge on the encoder side improves capacity.

Problem 2 *Symbol Error Rate vs. Message Error Rate*

We will prove the following theorem:

Theorem 1. (Converse Part of the Source-Channel Coding Theorem) *Let $V_1, \ldots, V_n$ be the outcome of an i.i.d. source of law P_V. Then, if $H(V) > C$ we have*

$$\frac{1}{n}\sum_{i=1}^{n} \Pr\Big[V_i \neq \hat{V}_i\Big] \not\to 0, \tag{7}$$

i.e. it is not possible to send the process over the channel with arbitrarily low probability of bit error.

1. We have to show that Theorem 1 implies: if $H(V) > C$ then

$$\Pr\Big[(V_1, V_2, \ldots, V_n) \neq (\hat{V}_1, \hat{V}_2, \ldots, \hat{V}_n)\Big] \not\to 0.$$

To this end, it suffices to prove that:

$$\begin{aligned}
\Pr\Big[(V_1, V_2, \ldots, V_n) \neq (\hat{V}_1, \hat{V}_2, \ldots, \hat{V}_n)\Big] &= \Pr\left[\bigcup_{i=1}^{n} V_i \neq \hat{V}_i\right] && (8)\\
&\geq \max_i \Pr\Big[V_i \neq \hat{V}_i\Big] && (9)\\
&\geq \frac{1}{n}\sum_{i=1}^{n} \Pr\Big[V_i \neq \hat{V}_i\Big], && (10)
\end{aligned}$$

where (9) follows because the union is a larger set, and (10) follows because the maximum of a set of numbers is always larger than the average of those set of numbers. Hence, if $\frac{1}{n}\sum_{i=1}^{n} \Pr\Big[V_i \neq \hat{V}_i\Big]$ does not tend to zero, then $\Pr\Big[(V_1, V_2, \ldots, V_n) \neq (\hat{V}_1, \hat{V}_2, \ldots, \hat{V}_n)\Big]$ can neither as it is always larger.

2. We state the proof again:

$$\begin{aligned}
\frac{1}{\mathsf{K}} H\left(U_1, \ldots, U_{\mathsf{K}} \middle| Z_1, \ldots, Z_{\mathsf{K}}\right) &= \frac{1}{\mathsf{K}}\sum_{i=1}^{\mathsf{K}} H\left(U_i \middle| U_1, \ldots, U_{i-1}, Z_1, \ldots, Z_{\mathsf{K}}\right) && (11)\\
&\leq \frac{1}{\mathsf{K}}\sum_{i=1}^{\mathsf{K}} H\left(U_i \middle| Z_1, \ldots, Z_{\mathsf{K}}\right) && (12)\\
&\leq \frac{1}{\mathsf{K}}\sum_{i=1}^{\mathsf{K}} H_{\mathrm{b}}(P_{\mathrm{e},i}) + \frac{1}{\mathsf{K}}\sum_{i=1}^{\mathsf{K}} P_{\mathrm{e},i}\log(a-1) && (13)\\
&\leq H_{\mathrm{b}}\left(\frac{1}{\mathsf{K}}\sum_{i=1}^{\mathsf{K}} P_{\mathrm{e},i}\right) + \log(a-1)\frac{1}{\mathsf{K}}\sum_{i=1}^{\mathsf{K}} P_{\mathrm{e},i} && (14)\\
&= H_{\mathrm{b}}(P_{\mathrm{b}}) + P_{\mathrm{b}}\log(a-1). && (15)
\end{aligned}$$

Equation (11) follows by the chain rule, (12) follows because conditioning reduces entropy, (13) follows by Fano's inequality, and (14) follows from Jensen's inequality and because entropy is concave.

3. Let V^n be the outcomes of an i.i.d source and $\hat{V}^n$ be the reconstruction sequence. Let X^n and Y^n be the corresponding inputs and outputs of the channel. We prove the converse to the source-channel separation theorem for memoryless channels (though it holds more generally.) Let C equal the capacity of the channel.

   Then

$$\begin{aligned}
H(V) &= \frac{1}{n}H(V_1, ..., V_n) && (16)\\
&= \frac{1}{n}\Big(H(V_1, ..., V_n) + H(V_1, \ldots, V_n|\hat{V}_1, \ldots, \hat{V}_n) \\
&\quad - H(V_1, \ldots, V_n|\hat{V}_1, \ldots, \hat{V}_n)\Big) && (17)\\
&= \frac{1}{n}H(V_1, \ldots, V_n|\hat{V}_1, \ldots, \hat{V}_n) + \frac{1}{n}I(V_1, \ldots, V_n; \hat{V}_1, \ldots, \hat{V}_n) && (18)\\
&\leq \frac{1}{n}H(V_1, \ldots, V_n|\hat{V}_1, \ldots, \hat{V}_n) + \frac{1}{n}I(X_1, \ldots, X_n; Y_1, \ldots, Y_n) && (19)\\
&\leq H_{\mathrm{b}}(P_{\mathrm{b}}) + P_{\mathrm{b}}\log(|\mathcal{V}| - 1) + \frac{1}{n}I\left(X_1, \ldots, X_n; Y_1, \ldots, Y_n\right) && (20)\\
&\leq H_{\mathrm{b}}(P_{\mathrm{b}}) + P_{\mathrm{b}}\log(|\mathcal{V}| - 1) + \frac{1}{n}nC && (21)\\
&= H_{\mathrm{b}}(P_{\mathrm{b}}) + P_{\mathrm{b}}\log(|\mathcal{V}| - 1) + C && (22)
\end{aligned}$$

   where Equation (16) follows from stationarity of the source, (19) follows by the data processing inequality because $\underline{V} \to \underline{X} \to \underline{Y} \to \underline{\hat{V}}$ forms a Markov chain, (20) follows from (15) with the definition

$$P_{\mathrm{b}} = \frac{1}{n}\sum_{i=1}^{n}\Pr\left[V_i \neq \hat{V}_i\right], \qquad (23)$$

   and (21) follows from the definition of capacity and because the channel is memoryless. If $P_{\mathrm{b}} \to 0$, then

$$H(V) \leq C. \qquad (24)$$

   We have proven now the following statement: if $P_{\mathrm{b}} \to 0$ then $H(V) \leq C$. Therefore, its contrapositive statement (which states: if $H(V) > C$ then $P_{\mathrm{b}} \not\to 0$) is also true and hence Theorem 1 is proved for memoryless channels.

Problem 3 — *Rate Distortion Function with Infinite Distortion*

First, we note that for the computation of the rate distortion function we only need to consider $D \in [0, 1/2]$, since for rate $R = 0$ we can achieve the distortion $D = 1/2$ by having $\hat{X} = 0$ with probability 1. Note also that achieving finite distortion requires $P_{\hat{X}|X}(\hat{x} = 0|x = 1) = 0$. Hence, the minimization of $I(X; \hat{X})$ can be restricted to conditional laws of the form

$$\begin{aligned}
&P_{\hat{X}|X}(\hat{x} = 0|x = 0) = 1 \qquad && P_{\hat{X}|X}(\hat{x} = 1|x = 0) = 0\\
&P_{\hat{X}|X}(\hat{x} = 0|x = 1) = p \qquad && P_{\hat{X}|X}(\hat{x} = 1|x = 1) = 1 - p,
\end{aligned}$$

with $p \in [0,1]$ satisfying $\mathsf{E}\left[d(X,\hat{X})\right] \leq D$. But the expected distortion relates to the probability p like

$$\begin{aligned}\mathsf{E}\left[d(X,\hat{X})\right] &= \Pr[X=1]\,\mathsf{E}\left[d(X,\hat{X}) \,\middle|\, X=1\right] \\ &= \frac{p}{2}.\end{aligned}$$

Hence, $\mathsf{E}\left[d(X,\hat{X})\right] \leq D$ is achieved if and only if $p \in [0,2D]$. On the other hand, the mutual information $I(X;\hat{X})$ depends on p as

$$\begin{aligned}I(X;\hat{X}) &= H(X) - H(X|\hat{X}) \\ &= 1 - \Pr\left[\hat{X}=0\right] H(X|\hat{X}=0) \\ &= 1 - \frac{1}{2}(1+p)H_{\mathrm{b}}\left(\frac{1}{1+p}\right).\end{aligned}$$

Since $I(X;\hat{X})$ is monotonically decreasing in p (for $p \in [0,1]$), its minimum (under the given constraints) is achieved by $p = 2D$. The rate distortion function is thus

$$R(D) = 1 - \left(\frac{1}{2} + D\right) H_{\mathrm{b}}\left(\frac{1}{1+2D}\right), \quad D \in \left[0, \frac{1}{2}\right].$$

Problem 4 ***Properties of*** $R(D)$

We have the two rate distortion functions

$$R(D) = \min_{P_{\hat{X}|X}:\mathsf{E}[d(x,\hat{x})]\leq D} I(X;\hat{X}),$$

and

$$R'(D) = \min_{P_{\hat{X}|X}:\mathsf{E}[d'(x,\hat{x})]\leq D} I(X;\hat{X}),$$

where the distortion measures $d(\cdot,\cdot)$ and $d'(\cdot,\cdot)$ relate through $d'(i,j) = d(i,j) - w_i$. We will show that $R'(D) = R(D+\bar{w})$ by expressing $\mathsf{E}\left[d'(X,\hat{X})\right]$ in terms of $\mathsf{E}\left[d(X,\hat{X})\right]$:

$$\begin{aligned}\mathsf{E}\left[d'(X,\hat{X})\right] &= \sum_{i=1}^{m}\sum_{j=1}^{m} p(i,j)\left(d(i,j) - w_i\right) \\ &= \sum_{i=1}^{m}\sum_{j=1}^{m} p(i,j)d(i,j) - \sum_{i=1}^{m}\sum_{j=1}^{m} p(i,j)w_i \\ &= \mathsf{E}\left[d(X,\hat{X})\right] - \sum_{i=1}^{m} p_i w_i \underbrace{\sum_{j=1}^{m} p(j|i)}_{=1} \\ &= \mathsf{E}\left[d(X,\hat{X})\right] - \bar{w}.\end{aligned}$$

Hence, we have

$$\begin{aligned}
R'(D) &= \min_{P_{\hat{X}|X}:\mathsf{E}\left[d'(X,\hat{X})\right]\le D} I(X;\hat{X}) \\
&= \min_{P_{\hat{X}|X}:\mathsf{E}\left[d(X,\hat{X})\right]-\bar{w}\le D} I(X;\hat{X}) \\
&= \min_{P_{\hat{X}|X}:\mathsf{E}\left[d(X,\hat{X})\right]\le D+\bar{w}} I(X;\hat{X}) \\
&= R(D+\bar{w}).
\end{aligned}$$

To conclude that there is no loss in generality in assuming that $\min_j d(i,j) = 0$ for all $i \in \{1,2,\ldots,m\}$, it remains to note that for every distortion measure $d(\cdot,\cdot)$, there exists a corresponding distortion measure $d'(\cdot,\cdot)$ with $\min_j d'(i,j) = 0$ for every $i \in \{1,2,\ldots,m\}$ (by choosing $w_i = \min_j d(i,j)$). And as we have shown, the rate distortion function $R(\cdot)$ then follows directly from $R'(\cdot)$ by

$$R(D) = R'(D-\bar{w}).$$